



STRUCTURED NOTES

CAPTURE • ORGANIZE • UNDERSTAND • APPLY

Name Andrea Febles
Course / Subject P.M.
Speaker / Class Carlos Pichardo
Date & Time 9-10-2026

KEYWORDS & KEY CONCEPTS

- Base 10
- dígitos
- potencias
-
-
-



Focus on the key concepts that generate value and understanding.

NOTES

Title: Matemáticas para la computación

Topic: Sistema decimal

Es el sistema que utilizamos habitualmente y tiene base 10, porque emplea diez dígitos, del 0 al 9. El valor de cada cifra depende de su posición y de las potencias de 10 que le corresponden.

ACTIONS / NEXT STEPS

- ☐
- ☐
- ☐
- ☐
- ☐

Turn knowledge into action.
Small steps, big results.

QUESTIONS & REFLECTIONS

¿Por qué el valor de una cifra cambia según su posición?



What questions do I have?



What is unclear or confusing?



What ideas connect?



How can I apply this?



What assumptions should I challenge?



SUMMARY

Write a brief summary of the main points.

Comprender este sistema ayuda a entender los demás sistemas



KEY TAKEAWAY

Capture the most important insight.

Carlos Pichardo Vinque

STRUCTURE NOTES V4 2026

Page: 1 of 11



KEEP LEARNING
KEEP GROWING
CREATE IMPACT



STRUCTURED NOTES

CAPTURE • ORGANIZE • UNDERSTAND • APPLY

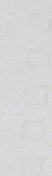
Name: Andrea Febles
Course / Subject: P.M
Speaker / Class: Carlos Pichardo
Date & Time: 9-10-2024

KEYWORDS & KEY CONCEPTS

- Base 2
- Bit
- cero
- uno
-
-

Focus on the key concepts that generate value and understanding.

ACTIONS / NEXT STEPS



Turn knowledge into action. Small steps, big results.

NOTES

Title: Sistemas numericos
Topic: Sistema binario

Es un sistema de base 2 que utiliza solamente los digitos 0 y 1.

Es fundamental para las computadoras, que representan y procesan la informacion mediante combinaciones de bits.

? QUESTIONS & REFLECTIONS

¿Que digitos usa el sistema binario?



What questions do I have?



What is unclear or confusing?



What ideas connect?



How can I apply this?



What assumptions should I challenge?



SUMMARY

Write a brief summary of the main points.

Este lenguaje es la base de la representacion digital



KEY TAKEAWAY

Capture the most important insight.

Carlos Pichardo Viague

• LIFELONG LEARNING •

STRUCTURE NOTES V4 2026

Page: 2 of 11



KEEP LEARNING
KEEP GROWING
CREATE IMPACT



STRUCTURED NOTES

CAPTURE • ORGANIZE • UNDERSTAND • APPLY

Name Andrea Pebles
Course / Subject P.M
Speaker / Class Carlos Richarcho
Date & Time 9-10-2026

KEYWORDS & KEY CONCEPTS

- Base 8
- digitos
- conversion
-
-
-



Focus on the key concepts that generate value and understanding.



ACTIONS / NEXT STEPS

- ☐
- ☐
- ☐
- ☐
- ☐



Turn knowledge into action.
Small steps, big results.

NOTES

Title: sistemas numericos
Topic: sistema octal

Es un sistema de base 8 que utiliza los digitos del 0 al 7. Permite representar cadenas binarias extensas de una manera mas corta y sencilla.



QUESTIONS & REFLECTIONS

¿Porque facilita la representacion des sistemas binarios?



What questions do I have?



What is unclear or confusing?



What ideas connect?



How can I apply this?



What assumptions should I challenge?



SUMMARY

Write a brief summary of the main points.

Informacion mas comprimida
facilita las cadenas binarias



KEY TAKEAWAY

Capture the most important insight.

Carlos Richarcho Viague

• LIFELONG LEARNING •

STRUCTURE NOTES V4 2026

Page: 3 of 11



KEEP LEARNING
KEEP GROWING
CREATE IMPACT



STRUCTURED NOTES

CAPTURE • ORGANIZE • UNDERSTAND • APPLY

Name Andrea Fabelo
Course / Subject P.M.
Speaker / Class Carlos Pichardo
Date & Time 9-10-2026

KEYWORDS & KEY CONCEPTS

- Base 16
- letras
- dígitos
-
-
-



Focus on the key concepts that generate value and understanding.

ACTIONS / NEXT STEPS



Turn knowledge into action.
Small steps, big results.

NOTES

Title: Sistemas numéricos
Topic: Sistema Hexadecimal

Es un sistema de base 16 que utiliza los dígitos del 0 al 9 y las letras de la A a la F. Se utiliza para expresar información binaria de manera compacta y facilitar su interpretación.

QUESTIONS & REFLECTIONS

¿Qué símbolos se utilizan en el sistema hexadecimal?



What questions do I have?



What is unclear or confusing?



What ideas connect?



How can I apply this?



What assumptions should I challenge?



SUMMARY

Write a brief summary of the main points.

Este sistema ayuda a simplificar grandes cantidades de información.



KEY TAKEAWAY

Capture the most important insight.

Carlos Pichardo Viñue

• LIFELONG LEARNING •

STRUCTURE NOTES V4 2026

Page: 4 of 11



KEEP LEARNING
KEEP GROWING
CREATE IMPACT



STRUCTURED NOTES

CAPTURE • ORGANIZE • UNDERSTAND • APPLY

Name Andrea Sebles
Course / Subject P.M
Speaker / Class Carlos Pichardo
Date & Time 9-10-2026

KEYWORDS & KEY CONCEPTS

- conversion
- Base
- division
- potencias
-

 Focus on the key concepts that generate value and understanding.

ACTIONS / NEXT STEPS



 Turn knowledge into action.
Small steps, big results.

NOTES

Title: Sistemas numericos

Topic: Generalizacion de las conversiones

Este tema explica como convertir cantidades de un sistema a otro. El metodo general consiste en convertir primero la cantidad decimal y despues al sistema que queremos tener, usando potencias, divisiones y multiplicaciones segun corresponde.

QUESTIONS & REFLECTIONS

¿Cual es la recomendacion para convertir de binario a decimal?



What questions do I have?



What is unclear or confusing?



What ideas connect?



How can I apply this?



What assumptions should I challenge?



SUMMARY

Write a brief summary of the main points.

Esto ayuda a hacer cambios necesarios sin perder datos.



KEY TAKEAWAY

Capture the most important insight.



KEEP LEARNING
KEEP GROWING
CREATE IMPACT

STRUCTURE NOTES V4 2026

Page: 5 of 11

• LIFELONG LEARNING •

Carlos Pichardo Vinque



STRUCTURED NOTES

CAPTURE • ORGANIZE • UNDERSTAND • APPLY

Name Andrea Febles

Course / Subject PM

Speaker / Class Carlos Pichardo

Date & Time 9-10-2026

KEYWORDS & KEY CONCEPTS

- suma
- acarreo
- base
- columnas
-
-

Focus on the key concepts that generate value and understanding.



ACTIONS / NEXT STEPS

- ☐
- ☐
- ☐
- ☐
- ☐

Turn knowledge into action.
Small steps, big results.



NOTES

Title: Sistemas numericos

Topic: Suma

La suma se realiza columna por columna, teniendo en cuenta la base del sistema. Cuando el resultado de una columna o supera la base, se genera un acarreo hacia la siguiente columna.

QUESTIONS & REFLECTIONS

¿Cuando se produce un acarreo al sumar en un sistema?



What questions do I have?



What is unclear or confusing?



What ideas connect?



How can I apply this?



What assumptions should I challenge?



SUMMARY

Write a brief summary of the main points.

Aprender el acarreo es importante para sumar bien



KEY TAKEAWAY

Capture the most important insight.

Carlos Pichardo Viagac

STRUCTURE NOTES V4 2026

• LIFELONG LEARNING •

Page: 6 of 11



KEEP LEARNING
KEEP GROWING
CREATE IMPACT



STRUCTURED NOTES

CAPTURE • ORGANIZE • UNDERSTAND • APPLY

Name Andrea Eablos
Course / Subject P.M.
Speaker / Class Carlos Pichardo
Date & Time 9-10-2026

KEYWORDS & KEY CONCEPTS

- Resta
- diferencia
- prestamo
- base
-
-

Focus on the key concepts that generate value and understanding.

ACTIONS / NEXT STEPS

- ☒
- ☐
- ☐
- ☐
- ☐

Turn knowledge into action.
Small steps, big results.

NOTES

Title: Sistemas numericos
Topic: Operaciones basicas: Resta

La resta permite hallar la diferencia entre dos cantidades y se efectua respetando los digitos validos de cada sistema. Cuando una cifra no alcanza para realizar la resta, se realiza el procedimiento de prestamo correspondiente a la base

QUESTIONS & REFLECTIONS

¿Cuando se utiliza el prestamo?



What questions do I have?



What is unclear or confusing?



What ideas connect?



How can I apply this?



What assumptions should I challenge?



SUMMARY

Write a brief summary of the main points

Usar correctamente el prestamo y sacar los buenos resultados



KEY TAKEAWAY

Capture the most important insight.

Carlos Pichardo Viñuales

STRUCTURE NOTES V4 2026

LIFELONG LEARNING

Page: 7 of 11



KEEP LEARNING
KEEP GROWING
CREATE IMPACT



STRUCTURED NOTES

CAPTURE • ORGANIZE • UNDERSTAND • APPLY

Name Andrea Febles
Course / Subject P.M
Speaker / Class Carlos Pichardo
Date & Time 9-10-2024

KEYWORDS & KEY CONCEPTS

- producto
- multiplicacion
- base
- producto
-



Focus on the key concepts that generate value and understanding.

ACTIONS / NEXT STEPS

- ☐
- ☐
- ☐
- ☐
- ☐



Turn knowledge into action.
Small steps, big results.

NOTES

Title: sistemas numericos

Topic: Multiplicacion

La multiplicacion se realiza mediante productos parciales y sumas, respetando las bases de los numeros.

Los resultados de cada operacion deben expresarse con digitos validos para el sistema utilizado.

QUESTIONS & REFLECTIONS



What questions do I have?



What is unclear or confusing?



What ideas connect?



How can I apply this?



What assumptions should I challenge?



SUMMARY

Write a brief summary of the main points.



KEY TAKEAWAY

Capture the most important insight.

Carlos Pichardo Viñue

• LIFELONG LEARNING •

STRUCTURE NOTES V4 2026

Page: 8 of 11



KEEP LEARNING
KEEP GROWING
CREATE IMPACT



STRUCTURED NOTES

CAPTURE • ORGANIZE • UNDERSTAND • APPLY

Name Andrea Febles
Course / Subject P.M
Speaker / Class Carlos Pichardo
Date & Time 9-10-2026

KEYWORDS & KEY CONCEPTS

- Division
- cociente
- residuo
- Base
-
-



Focus on the key concepts that generate value and understanding.

ACTIONS / NEXT STEPS

- ☐
- ☐
- ☐
- ☐
- ☐

Turn knowledge into action.
Small steps, big results.

NOTES

Title: Sistemas numericos

Topic: Division

La division permite determinar cuantas veces una cantidad esta contenida en otra.

Al realizarla en un sistema numerico, se deben respetar sus reglas y expresar el cociente y el residuo de acuerdo con la base correspondiente.

QUESTIONS & REFLECTIONS

¿Cuales son las partes de una division?



What questions do I have?



What is unclear or confusing?



What ideas connect?



How can I apply this?



What assumptions should I challenge?



SUMMARY

Write a brief summary of the main points.

La division es importante para la conversion de sistemas



KEY TAKEAWAY

Capture the most important insight.

Carlos Pichardo Vinque

LIFELONG LEARNING

STRUCTURE NOTES V4 2026

Page: 9 of 11



KEEP LEARN
KEEP GROW
CREATE IN



STRUCTURED NOTES

CAPTURE • ORGANIZE • UNDERSTAND • APPLY

Name Andrea Febles
Course / Subject P.M.
Speaker / Class Carlos Pichardo
Date & Time 9-10-2026

KEYWORDS & KEY CONCEPTS

- Binario
- complemento a 2
- bit de signo
-
-
-



Focus on the key concepts that generate value and understanding.

ACTIONS / NEXT STEPS

- ☐
- ☐
- ☐
- ☐
- ☐



Turn knowledge into action.
Small steps, big results.

NOTES

Title: Sistemas numericos

Topic: complemento a 2

El complemento a 2 es una tecnica utilizada para representar numeros negativos y realizar sumas binarias que equivalen a restas. Para obtenerlo, se invierten los bits de la cantidad y se suma uno, siguiendo la representacion empleada en el capitulo.

QUESTIONS & REFLECTIONS

¿Como se obtiene el complemento a 2 de un numero?



What questions do I have?



What is unclear or confusing?



What ideas connect?



How can I apply this?



What assumptions should I challenge?



SUMMARY

Write a brief summary of the main points.

El complemento a 2 ayuda a trabajar con binarios negativos



KEY TAKEAWAY

Capture the most important insight.



KEEP LEARN
KEEP GROW
CREATE IM

STRUCTURE NOTES V4 2026

Page: 10 of 11

• LIFELONG LEARNING •

Carlos Pichardo Vinque



STRUCTURED NOTES

CAPTURE • ORGANIZE • UNDERSTAND • APPLY

Name Andrea Febles
Course / Subject P. M
Speaker / Class Carlos Pichardo
Date & Time 9-10-2026

KEYWORDS & KEY CONCEPTS

- computadora
- binario
- ASCII
- informacion
-

Focus on the key concepts that generate value and understanding.

ACTIONS / NEXT STEPS

- ☐
- ☐
- ☐
- ☐
- ☐

Turn knowledge into action.
Small steps, big results.

NOTES

Title: sistemas numericos

Topic: Aplicaciones

Los sistemas numericos permiten representar y procesar informacion dentro de la computadoras. El binario es fundamental para las operaciones internas, mientras que el octal, hexadecimal y codigos como ASCII ayudan a representar o interpretar informacion de forma mas practica.

QUESTIONS & REFLECTIONS

¿Porque las computadoras usan el sistema binario?

What questions do I have?

What is unclear or confusing?

What ideas connect?

How can I apply this?

What assumptions should I challenge?



SUMMARY

Write a brief summary of the main points.



KEY TAKEAWAY

Capture the most important insight.

Carlos Pichardo Vique

• LIFELONG LEARNING •

STRUCTURE NOTES V4 2026

Page: 11 of 11



KEEP LEARNING
KEEP GROWING
CREATE IMPACT